

Are grasslands changing gears: Long-term global grassland experiment reveals a shift in C₄ vegetation cover driven by temperature and soil nutrient availability

Ezinwanne Ezekannagha; Nicholas Smith
Department of Biological Science, Texas Tech University, Lubbock, Tx

Grasslands are important land carbon sinks, helping to mitigate the increase in atmospheric carbon dioxide (CO₂). Carbon sequestration in grasslands is initiated by photosynthesis, wherein plants assimilate carbon that is essential for sustaining and maintaining plant tissues as well as supporting growth. Within grasslands, plants perform photosynthesis differently using either C₃ or C₄ photosynthetic pathways. The prevalence of C₃ and C₄ plants is influenced by factors such as temperature, aridity, and soil nutrients. C₄ plants thrive in hot, dry, and water-limited environments due to their efficient CO₂ fixation, while C₃ plants thrive in cool, wet conditions. Warming temperatures typically favor C₄ plants due to their lower photorespiration, which increases with temperature and reduces C₃ photosynthesis. In addition to temperature, low water, and nutrient availability further tend to favor C₄ plants due to their higher water and nutrient use efficiency. Nonetheless, despite these well-established predictions, there remains limited exploration of these dynamics at large spatial scales, leaving uncertainties regarding the interaction of environmental factors in influencing the relative cover of C₃ and C₄ plants.

Here, we quantified the direct and interactive effects of temperature, aridity, and soil nutrient availability on C₃ and C₄ grass vegetation cover using 16 years (2007 to 2023) of data from a global, coordinated nutrient addition experiment spanning 171 grassland sites. Our findings revealed a strong positive effect of annual growing season temperature and a negative effect of soil nutrient availability on the relative C₄ vegetation cover. Surprisingly, no significant impact of aridity on the C₄ cover was observed, challenging previous assumptions. These results not only support existing hypotheses regarding the impact of temperature and soil nutrients but also present a nuanced perspective on the role of aridity. Furthermore, the results suggest a potential shift in the relative cover of grasslands favoring the C₄ pathway with future global warming, although this may be balanced by eutrophication. Our findings provide benchmarks for predicting future changes in grassland C₃ and C₄ vegetation cover that can be used to better estimate future carbon sequestration in these vital ecosystems.